Yuheng(Jeffery) Fang

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Education

Binghamton University — **January 2025 – Present**
Master of Science in Computer Science; GPA: 3.62/4 — *Binghamton, New York*

Rensselaer Polytechnic Institute — **Sep. 2019 – May 2023**
Bachelor of Science in Biomedical Engineering and Minor in Psychology — *Troy, New York*

Experience

Montefiore Medical Center — **July 2023 – August 2024**
Techologist — *Bronx, New York*

- Prepared approximately 80 samples consisting of blood, bone marrow, tissue, cerebrospinal fluid, and stem cells a day to diagnose diseases through the usage of flow cytometry for further patient care.
- Stained lymphocyte subsets using techniques consisting of micropipettor, hemacytometer, cytocentrifuge, and more for Beckman Coulter Aquios and Navios instrument for further analysis.
- Analyzed and gated flow cytometry data for over 15 different panels using the Kaluza software which consists of diseases such as acute leukemia and paroxysmal nocturnal hemoglobinuria.
- Operated, maintained, and troubleshoot flow cytometry machines for precise laser alignment and laminar flow of sheath fluid to ensure functioning machine as well as passing quality control tests for accurate results.

Eye Care 2000 — **August 2021 – December 2021**
Optometrist Assistant — *Houston, Texas*

- Operated diagnostic equipment such as autorefractors, tonometers, and lensometers to perform pre-examination tests under supervision.
- Managed appointment scheduling, patient records, and insurance documentation, ensuring accuracy and confidentiality.
- Gained insight into the daily operations of an optometry practice, including the importance of patient care, precise communication, and teamwork in a healthcare setting.
- Assisted with patient intake, including gathering medical history, conducting preliminary vision tests, and preparing patients for examinations.

Projects

Senior-Targeted Intelligent Assessment — **January 2025 – August 2026**

- Structured the end-to-end system integrating hardware (biosensor reagent card), computer vision/image analysis, and cloud infrastructure, achieving 99%+ diagnostic accuracy validated against third-party lab results
- Built a cloud-based data pipeline for point-of-care testing (POCT) devices: automated data acquisition, standardization, and storage into a real-time pathological database with anomaly detection for abnormal results
- Led a cross-functional, transnational engineering and clinical team (U.S. + China) from concept through third-party validation trials.

Multi-functional Surgical Device — **August 2022 – May 2023**

- Participated in product research, benchmarking, creation of prototyping, testing, and writing of technical reports to support the development of the medical device while adhering to FDA guidelines for potential future product release.
- Developed a laparoscopic surgical device with the functions of both cutting and applying clips with minimal time and momentum lost
- Utilized SolidWorks to construct different CAD and 3D printed each design to test.

Machine Learning of Vessel Segmentation — **January 2023 – May 2023**

- Through rigorous experimentation using 3 very different deep learning-based models, establish the ability to circumvent transfer learning and domain adaptation-based approaches while doing vessel segmentation on retinal scans
- Using the programming language Python, two team members and I developed a 70-30 split from each of the datasets and created the extended train-test split comprising samples from all the 3 datasets.

Technical Skills

Languages: Python, Java, C, C++, C#, SQL, Matlab
Software: VS Code, Labview, Minitab, NX, Solidworks, Softmax Pro, Microsoft Office, Kaluza, MongoDB
Lab Techniques: Aseptic Techniques, Biofire Mycoplasma, CBC-Diff, Cell Culture, Cell Staining, Flow Cytometry, Instron, Liquid Chromatography, Molecular Biology Techniques, Hemacytometer, Pelleting, Pipetting, Spectroscopy